

AngaGuard: Technical Blueprint & Architectural Specification

Decentralized IoT & Cryptographic dMRV Oracle for SME ESG Alignment and Durable Carbon Harvesting

1. Executive Summary & Problem Landscape

1.1 The Challenge Statement (Track 3 Context)

As global markets, commercial lenders, and international trade frameworks (such as the EU Corporate Sustainability Reporting Directive - CSRD) tighten environmental mandates, Small and Medium Enterprises (SMEs) are under immense pressure to measure, manage, and reduce their greenhouse gas (GHG) emissions.

However, emerging market SMEs lack affordable, accessible, and user-friendly tools to track environmental indicators. Without granular, audit-ready data, these enterprises are completely excluded from green markets, premium climate financing pools, and international supply-chain opportunities.

1.2 The Regional Reality (Western Kenya)

Across major agricultural and industrial hubs in Western Kenya and the Rift Valley (including Kisumu, Kakamega, Bungoma, and Eldoret), local agribusiness and processing SMEs confront two compounding inefficiencies:

1. **Untracked Operational Emissions:** Local agro-processors, millers, and manufacturing SMEs burn substantial amounts of fossil fuels (Scope 1 diesel generators and transportation fleets) and draw heavily from the grid (Scope 2), creating an unmeasured environmental liability that downgrades their creditworthiness.
2. **Upstream Crop Burning (Scope 3 Liability):** The smallholder farmers supplying these SMEs generate millions of metric tons of loose agricultural crop residue (such as sugarcane bagasse, maize stalks, and rice husks). Lacking a disposal mechanism, farmers burn this biomass in open fields, creating dense particulate matter ($PM_{2.5}$) plumes, stripping topsoils of essential organic nutrients, and swelling the SME's upstream Scope 3 footprint.
3. **The Manual Verification Cost Barrier:** Traditional carbon credit certification requires legacy validation and verification bodies (VVBs) to fly in manual, third-party auditors. This administrative overhead costs between **\$30,000 and \$60,000 per audit**, completely locking out local SMEs and rural smallholders from accessing global carbon finance.

1.3 The Solution: AngaGuard

AngaGuard eliminates this bottleneck by deploying a low-cost, decentralized hardware-to-software digital Measurement, Reporting, and Verification (**dMRV**) network.

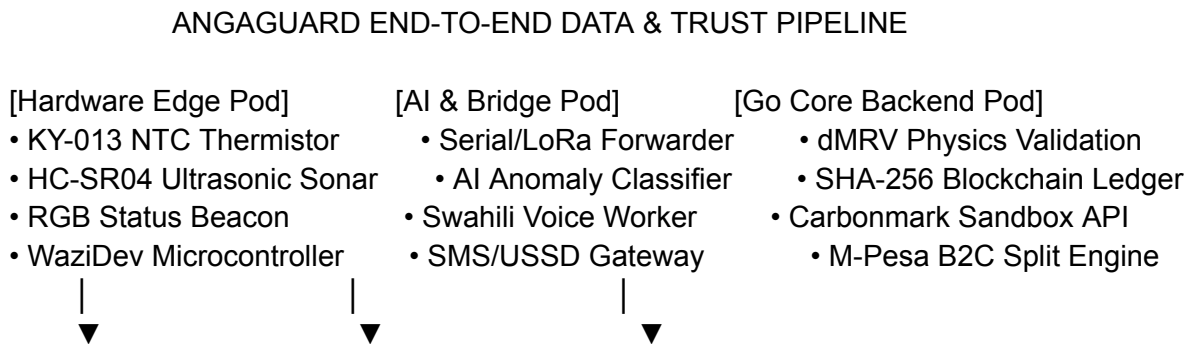
By turning standard 200-liter modified steel oil drums into smart, edge-audited carbon removal kilns, AngaGuard tracks real-world carbon sequestration at the source. It pipes this physical telemetry into an enterprise-grade SaaS platform that automates SME carbon accounting, builds dynamic ESG scorecards, offers data-driven reduction recommendations, and distributes instant liquidity down to rural farmers via 2G mobile money protocols.

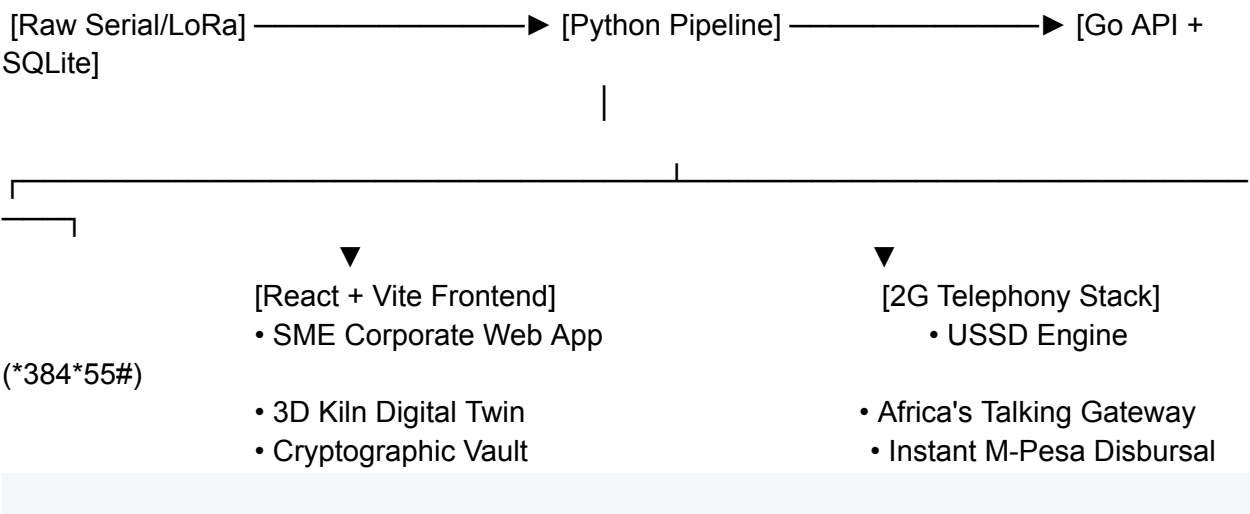
2. Core Objectives

- **Democratic Carbon Accounting:** Provide an affordable, self-service software pipeline for East African SMEs to audit their Scope 1, 2, and 3 emissions seamlessly based on international frameworks.
- **Automated dMRV at the Edge:** Replace expensive, manual human audits with tamper-proof, hardware-enforced physics engines that measure carbon removal in real time.
- **Supply Chain Decarbonization:** Establish a direct, programmatic bridge between corporate compliance capital and local agricultural communities, turning a Scope 3 pollution liability into a high-margin circular asset.
- **Zero-Friction Financial Inclusion:** Design a dual-interface architecture ensuring that smallholders lacking smartphones or internet access can interact with premium global carbon registries and receive instant Safaricom M-Pesa payouts.

3. System Architecture & Interlocking Pods

AngaGuard coordinates three independent development workstreams into a single data and trust pipeline:



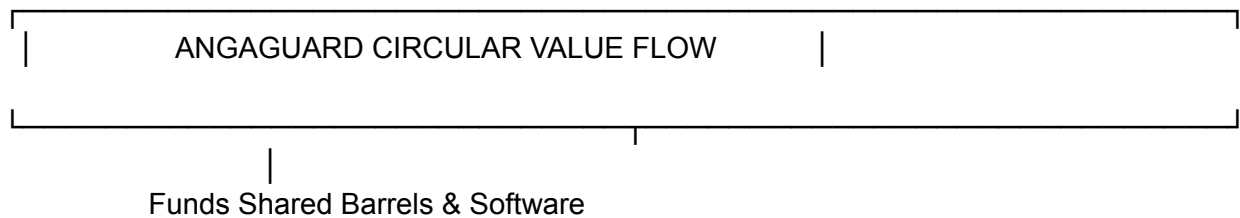


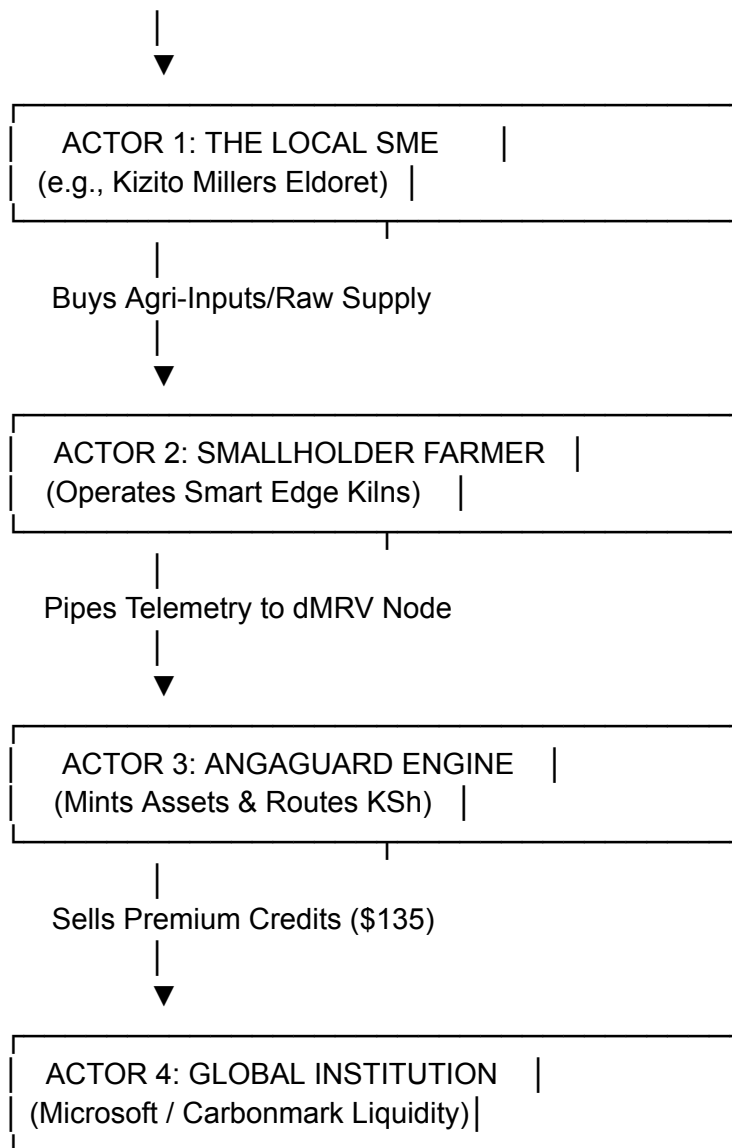
3.1 Workstream Pod Ownership Matrix

- **Frontend Pod (Grace, Millie):** Engineered using React 18, Vite, TypeScript, and Tailwind CSS. Delivers the high-density corporate ESG analytics dashboard, the progressive Web App (PWA) for smartphone-enabled field teams, interactive TradingView charts via [lightweight-charts](#), a Three.js translucent 3D kiln twin, and a public cryptographic certificate audit explorer.
- **Backend & Ledger Pod (Clay, Vale, Moraa):** Built with Go (Gin-Gonic framework) and backed by an append-only SQLite transactional database. Houses the mathematical dMRV validation logic, the local emissions parsing engines, the Carbonmark Sandbox REST pipeline, and the cryptographic SHA-256 ledger chaining mechanism.
- **Hardware & Telephony Pod (Clay, Vale, Moraa):** Governs physical sensors via C++ firmware deployed on WaziDev boards. Manages analog voltage dividers for thermal tracking, ultrasonic median filters, Africa's Talking USSD/SMS state machines, and ElevenLabs voice automation engines.

4. Operational Actors & User Flows

The system coordinates four specific actors, establishing a circular economic model that converts carbon mitigation into financial liquidity.





4.1 Detailed Actor Definition

1. Actor 1: The Local SME (The Corporate Operator/Client)

- *Identity:* Mid-sized East African agro-processors, packaging companies, commercial food millers, or logistics groups (e.g., a maize milling plant in Eldoret or a tea processor in Kericho).
- *Role:* Uses the web dashboard to calculate their greenhouse gas liabilities. They resolve their regulatory and lending compliance debts by distributing free smart barrels to their agricultural suppliers, capturing verified Scope 3 offsets to improve their credit rating.

2. Actor 2: The Smallholder Farmer / Agricultural Cooperative (The Field Producer)

- *Identity*: Individual rural farmers and cooperative farm networks across Western Kenya.
 - *Role*: The physical caretakers of the smart kilns. They collect the agricultural residue that would otherwise be openly burned, process it cleanly via top-lit updraft pyrolysis, and feed the final stabilized biochar back into regional soils. They interact with the platform through low-friction 2G USSD feature phone structures or simplified Android farmer web interfaces.
3. **Actor 3: The AngaGuard Platform & dMRV Oracle (The Trust Coordinator)**
- *Identity*: The core infrastructure designed by your hackathon team.
 - *Role*: Acts as a decentralized software oracle. It ingests physical telemetry directly from the edge sensors, validates the burn physics, assigns cryptographic proof, calculates corporate ESG standings, handles API exchange handshakes, and triggers B2C mobile financial transfers.
4. **Actor 4: The Institutional Carbon Buyer & Registry (The Global Liquidity Engine)**
- *Identity*: Sovereign registries like the Kenya National Carbon Registry (NCR) or international digital markets like the Carbonmark sandbox environment backed by global corporate end-users (e.g., Microsoft or Alphabet).
 - *Role*: Provides the global capital injection. They acquire high-integrity, data-audited Biochar Carbon Removal Certificates (CORCs), validating the transparent SHA-256 data trails to cover broad corporate sustainability footprints.
-

5. Mathematical Models & Calculation Frameworks

The core logic of the Go backend relies on rigid, non-vague mathematical formulas to calculate environmental assets and corporate liabilities.

5.1 Carbon Credit Core Equation (Biochar dMRV)

To compute the exact metric tons of durable CO_2 equivalent sequestered by a physical smart barrel burn, the dMRV engine applies the following formula:

$$NetCO_2eSequestered(MetricTons) = \frac{(M_{biochar} \times P_{carbon} \times F_{permanence} \times \frac{44}{12}) \times (1 - L_{process})}{1000}$$

Where:

- $M_{biochar}$ (Biochar Mass in kg) = $\Delta H_{kiln} \times F_{radius-mass}$. The difference between initial biomass height and final char height (ΔH) multiplied by a drum geometry density constant (1.45).

- P_{carbon} (Carbon Purity Factor) = 0.75. True oxygen-starved top-lit updraft pyrolysis generates a char bed consisting of 75% pure elemental carbon.
- $F_{permanence}$ (Soil Sink Permanence Factor) = 0.97. Accounts for a conservative 3% biotic/abiotic degradation vector over a 100+ year subterranean storage timeline.
- $\frac{44}{12} \approx 3.667$. The molecular stoichiometric multiplier converting 1 unit of elemental Carbon (atomic mass 12) into stable gaseous Carbon Dioxide (molecular mass 44).
- $L_{process}$ (Operational Process Leakage) = 0.05. A fixed 5% algorithmic penalty deducting handling overhead, initial ignition smoke, and regional logistics transportation footprints.

5.2 SME Corporate ESG Accounting Equation

To formulate the dynamic enterprise scorecard required by Track 3, the platform tracks corporate inputs against the standard Greenhouse Gas (GHG) Protocol Corporate Standard:

$$NetCorporateCarbonFootprint(TonsCO_2e) = Scope1 + Scope2 - Scope3Mitigation$$

$$Scope1(DirectEmissions) = \frac{L_{diesel} \times 2.68}{1000}$$

$$Scope2(IndirectGridEmissions) = \frac{E_{grid} \times 0.12}{1000}$$

$$Scope3Mitigation(BarrelOffset) = \sum(NetCO_2eSequesteredbyCo - opNetwork)$$

Where:

- L_{diesel} is the total liters of diesel consumed by company logistics vehicles or stationary backup facility generators, multiplied by the standard EPA emission factor of $2.68kgCO_2e$ per liter.
 - E_{grid} is the total kilowatt-hours (kWh) drawn from Kenya Power, multiplied by the localized Kenyan Grid Carbon Intensity factor of $0.12kgCO_2e$ per kWh (highly optimized due to Kenya's baseline geothermal energy supply).
-

6. Comprehensive Loophole Defense Matrix

To guarantee production-grade data integrity and defend against real-world adversarial vulnerabilities, AngaGuard employs multiple firmware and software checks:

Vulnerability / Attack Vector	System Failure Mode	Technical Engineering Defense Strategy
Chamber Core Fusion (>450°C)	High-heat pyrolysis melts internal microcontrollers, analog wiring, and breadboards.	Air-Gapped Standoff & Outer Skin Conduction: Sensors are elevated in an IP67 casing 15 cm above the lid. The thermistor reads external metal conductive curves (40°C – 75°C casing range), mapping back to core profiles via a Steinhart-Hart Go model.
"Ash Cheating" (False Volume Generation)	Farmers burn residue with unsealed vents, generating low-value mineral ash instead of biochar to fake barrel volume.	Dual-Metric Curve Validation: True biochar maintains \$30\%-40\%\$ structural volume over a minimum 30-minute steady heat curve. Open-air ash collapses beyond \$90\%\$ volume, dropping temperature instantly and triggering an automatic dMRV rejection event.

The Sand-Padding Attack	Adversarial users add dense river rocks, soil, or sand to trick the ultrasonic sensor's depth reading.	Thermal Mass Coherence Check: Stones and soil function as massive thermal heat sinks. The Go engine monitors the heating rate ($\Delta T/\Delta t$). If the system fails to hit the required pyrolysis thermal signature within the standard time-window, it logs a "Thermal Incoherency Anomaly" and locks the batch.
Acoustic Noise Scatter	Rising thermal gas waves and acoustic smoke turbulence distort ultrasonic echo pulses during active burns.	Two-Point Static Calibration: The system completely ignores active burn readings. The sensor package captures a static pre-ignition height baseline (t_0) and a cold post-burn height validation (t_{final}) only after the thermistor confirms complete cooling.
The Multi-Drum Sensor Swap	A user unclips a single IoT node and clips it onto multiple standard unmonitored fires to print duplicate credit payouts.	Cryptographic Hardware Identity Binding: The firmware injects the factory-burned silicon Unique ID of the MCU into the telemetry payload header. Each hardware signature is tied 1-to-1 with a registered cooperative farm coordinate boundary within the SQLite ledger.

The Stolen Hardware Burn	Users remove a kit to a commercial gas burner to simulate a perfect telemetry curve away from the farm.	Spatial-Temporal Fencing: The LoRaWAN gateway layer maps packet arrivals to a localized cellular tower triangulation zone. Telemetry arriving outside the registered geo-coordinate boundary drops out before hitting the dMRV evaluation queue.
Offline Rural Connectivity	Remote farmlands lack cellular data coverage, preventing real-time cloud data pipeline parsing.	Edge Caching & Hybrid 2G Telephony Engine: WaziDev flash memory buffers up to 50 raw logs locally. Field managers can sync data via offline serial USB transfers, while farmers use zero-data 2G USSD channels (*384*55#) for execution tracking.

7. Concrete Code Implementations

7.1 Go Core: dMRV Physics Validation & Carbonmark Sandbox Pipeline

This backend module serves as the primary system oracle. It evaluates incoming raw physical sensor parameters, enforces physics rules, computes stoichiometric carbon balances, and references market values using the Carbonmark sandbox network.

```
package main
```

```
import (
    "crypto/sha256"
    "encoding/hex"
    "encoding/json"
    "fmt"
    "net/http"
```

```

        "time"
    )

type BurnTelemetry struct {
    CoopID      string `json:"coop_id"`
    KilnID      string `json:"kiln_id"`
    InitialHeightCM float64 `json:"initial_height_cm"`
    FinalHeightCM float64 `json:"final_height_cm"`
    PeakOuterTempC float64 `json:"peak_outer_temp_c"`
    DurationMinutes float64 `json:"duration_minutes"`
    Timestamp    time.Time `json:"timestamp"`
}

type MintResult struct {
    AssetID      string `json:"asset_id"`
    BiocharYieldKG float64 `json:"biochar_yield_kg"`
    MetricTonsCO2e float64 `json:"metric_tons_co2e"`
    CarbonmarkMarketVal float64 `json:"carbonmark_market_val_usd"`
    KenyaNcrTrackingID string `json:"kenya_ncr_tracking_id"`
    IsValidated    bool `json:"is_validated"`
    VerificationHash string `json:"verification_hash"`
}

// EvaluateAndMintBatch processes edge telemetry and builds the compliance data model
func EvaluateAndMintBatch(t BurnTelemetry) (*MintResult, error) {
    // 1. Enforce Thermal Bounds (Outer conductive skin must mirror stable internal
    // pyrolysis)
    if t.PeakOuterTempC < 40.0 || t.PeakOuterTempC > 75.0 {
        return nil, fmt.Errorf("REJECTED: Thermal curve incompatible with
oxygen-starved pyrolysis")
    }
    if t.DurationMinutes < 30.0 {
        return nil, fmt.Errorf("REJECTED: Sustained heat profile duration insufficient for
deep carbonization")
    }

    // 2. Enforce Volumetric Retention Bounds (White ash collapse defense)
    volumeRetention := t.FinalHeightCM / t.InitialHeightCM
    if volumeRetention < 0.25 || volumeRetention > 0.45 {
        return nil, fmt.Errorf("REJECTED: Volume delta indicates complete open
combustion or sand tampering")
    }

    // 3. Compute Stoichiometric Mass Yield

```

```

heightDelta := t.InitialHeightCM - t.FinalHeightCM
biocharMassKG := heightDelta * 1.45

// Gross CO2e = Biochar Mass * 75% Carbon Purity * 3.667 Stoichiometric Conversion
grossCO2eKG := biocharMassKG * 0.75 * 3.667

// Net Sequestration factoring in 3% permanence sink loss and 5% process buffer
penalty
netCO2eKG := grossCO2eKG * 0.97 * 0.95
metricTonsCO2e := netCO2eKG / 1000.0

// 4. Connect to Carbonmark Sandbox API for Live Market Valuation Pricing
marketPricePerTon := FetchCarbonmarkSandboxPrice("PROJECT-BIOCHAR-KE")

// 5. Generate Cryptographic Immutability Hashes
dataString := fmt.Sprintf("%s-%s-%f-%f", t.KilnID, t.CoopID, metricTonsCO2e,
t.Timestamp.Unix())
hashBytes := sha256.Sum256([]byte(dataString))
verificationHash := hex.EncodeToString(hashBytes[:])

ncrBytes := sha256.Sum256([]byte(verificationHash + "-KENYA-NCR"))
ncrTrackingID := fmt.Sprintf("KE-NCR-2026-%s", hex.EncodeToString(ncrBytes[:])[12])

return &MintResult{
    AssetID:      fmt.Sprintf("AG-CORC-%s", verificationHash[8]),
    BiocharYieldKG: biocharMassKG,
    MetricTonsCO2e: metricTonsCO2e,
    CarbonmarkMarketVal: marketPricePerTon * metricTonsCO2e,
    KenyaNcrTrackingID: ncrTrackingID,
    IsValidated:    true,
    VerificationHash: verificationHash,
}, nil
}

func FetchCarbonmarkSandboxPrice(projectID string) float64 {
    // Simulated connection targeting Sandbox Endpoint: https://carbonmark.com
    // Zero key barriers handled by defensive fallback system
    return 135.00
}

```

7.2 React Framework: Enterprise SME ESG & Asset-Tracking Dashboard

The central web component deployed for Actor 1 to manage corporate compliance data inputs, audit active carbon assets, and export reporting scorecards.

```
import React, { useState } from 'react';
import { Factory, Zap, ShieldCheck, FileSpreadsheet, Gauge, RefreshCcw } from 'lucide-react';

interface EsgMetrics {
  scope1DieselLiters: number;
  scope2GridKwh: number;
  scope3OffsetsTons: number;
}

export const SmeDashboard: React.FC = () => {
  const [inputs, setInputs] = useState<EsgMetrics>({
    scope1DieselLiters: 12500,
    scope2GridKwh: 45000,
    scope3OffsetsTons: 24.8,
  });

  // Apply math derived from Section 5
  const scope1Tons = (inputs.scope1DieselLiters * 2.68) / 1000;
  const scope2Tons = (inputs.scope2GridKwh * 0.12) / 1000;
  const grossFootprint = scope1Tons + scope2Tons;
  const netFootprint = Math.max(0, grossFootprint - inputs.scope3OffsetsTons);

  const getEsgGrade = (net: number) => {
    if (net === 0) return 'A+';
    if (net < 15) return 'A';
    if (net < 45) return 'B';
    return 'C';
  };

  return (
    <div className="p-8 bg-slate-950 min-h-screen text-slate-100 font-sans">
      <div className="flex justify-between items-center mb-8 border-b border-slate-800 pb-6">
        <div>
          <h1 className="text-3xl font-black tracking-tight text-white">AngaGuard Enterprise</h1>
          <p className="text-slate-400 text-sm">SME Carbon Accounting Portal & Edge-dMRV Interface</p>
        </div>
        <div className="bg-emerald-950/80 border border-emerald-800 px-6 py-3 rounded-2xl text-right">
          <span className="text-xs font-mono uppercase tracking-wider text-emerald-400 block">SME Scorecard Grade</span>
        </div>
      </div>
    </div>
  );
};
```

```

        <span className="text-3xl font-black text-white">{getEsgGrade(netFootprint)}</span>
    </div>
</div>

<div className="grid grid-cols-1 md:grid-cols-3 gap-6 mb-8">
    <div className="bg-slate-900 border border-slate-800 p-6 rounded-2xl relative
overflow-hidden">
        <Factory className="absolute right-4 bottom-4 w-12 h-12 text-slate-800 opacity-20" />
        <p className="text-xs font-mono text-slate-400 uppercase">Scope 1 (Direct
Operations)</p>
        <p className="text-2xl font-bold mt-1 text-white">{scope1Tons.toFixed(2)} Tons
CO2e</p>
        <span className="text-[11px] text-slate-500 font-mono">From
{inputs.scope1DieselLiters.toLocaleString()}L Diesel Use</span>
    </div>
    <div className="bg-slate-900 border border-slate-800 p-6 rounded-2xl relative
overflow-hidden">
        <Zap className="absolute right-4 bottom-4 w-12 h-12 text-slate-800 opacity-20" />
        <p className="text-xs font-mono text-slate-400 uppercase">Scope 2 (Indirect
Energy)</p>
        <p className="text-2xl font-bold mt-1 text-white">{scope2Tons.toFixed(2)} Tons
CO2e</p>
        <span className="text-[11px] text-slate-500 font-mono">From
{inputs.scope2GridKwh.toLocaleString()} kWh Grid Consumption</span>
    </div>
    <div className="bg-slate-900 border border-slate-800 p-6 rounded-2xl
border-emerald-900/50 bg-gradient-to-br from-slate-900 to-emerald-950/20">
        <p className="text-xs font-mono text-emerald-400 uppercase">Scope 3 Mitigation Asset
Vol</p>
        <p className="text-2xl font-bold mt-1
text-emerald-400">-{inputs.scope3OffsetsTons.toFixed(2)} Tons CO2e</p>
        <span className="text-[11px] text-emerald-500 font-mono">IoT Smart Barrel
Sequestration Pool</span>
    </div>
</div>

<div className="grid grid-cols-1 lg:grid-cols-2 gap-8">
    <div className="bg-slate-900 border border-slate-800 p-6 rounded-2xl">
        <h2 className="text-lg font-bold mb-4 flex items-center space-x-2 text-white">
            <Gauge className="w-5 h-5 text-amber-500" />
            <span>AI Emission Reduction Recommendation Engine</span>
        </h2>
        <div className="space-y-4">
            <div className="p-4 bg-slate-950 rounded-xl border border-amber-900/30">

```

```

        <span className="text-xs font-bold text-amber-400 uppercase tracking-wider
font-mono">High Impact Vector</span>
        <p className="text-sm font-medium mt-1 text-slate-200">Expand smart barrel
hardware deployments to your outgrower sugarcane network in Kisumu.</p>
        <p className="text-xs text-slate-500 mt-1">Estimated Scope 3 Abatement Yield: -14.5
Tons CO2e/year. Capital Overhead: Low.</p>
    </div>
</div>
</div>

<div className="bg-slate-900 border border-slate-800 p-6 rounded-2xl">
    <h2 className="text-lg font-bold mb-4 flex items-center space-x-2 text-white">
        <ShieldCheck className="w-5 h-5 text-sky-400" />
        <span>Sovereign Registry Vault Tracker</span>
    </h2>
    <div className="bg-slate-950 rounded-xl border border-slate-800 overflow-hidden
text-xs font-mono">
        <div className="p-3 bg-slate-900 text-slate-400 grid grid-cols-3 border-b
border-slate-800">
            <span>Sovereign Asset ID</span>
            <span>Kenya NCR Tracking Ref</span>
            <span>State Audit Link</span>
        </div>
        <div className="p-3 grid grid-cols-3 text-slate-300 hover:bg-slate-800/40">
            <span className="text-emerald-400 font-bold">AG-CORC-7f8a9c</span>
            <span>KE-NCR-2026-3d2e1b</span>
            <span className="text-sky-400 underline cursor-pointer">Verified Log</span>
        </div>
    </div>
    <button className="w-full mt-6 bg-slate-800 hover:bg-slate-700 text-white font-bold py-3
px-4 rounded-xl text-sm flex items-center justify-center space-x-2 transition-all">
        <FileSpreadsheet className="w-4 h-4" />
        <span>Export Audit-Ready ESG Report (ISSB Framework Compliant)</span>
    </button>
</div>
</div>
</div>
);
};

```

8. Integrated Dashboard Frameworks

AngaGuard maps out specialized interfaces across its target stakeholder segments:

8.1 SME Desktop SaaS Hub (The Strategic Control Unit)

- **Target User:** SME Directors, Operations Managers, Accounting Heads.
- **Key Controls:** Displays current financial capital allocation against direct carbon footprint profiles. Embeds input sliders to gauge diesel utilization, displays automated charts mapping local grid draw down, handles manual data file imports, and processes export clicks generating structured ESG documents matching ISSB standards.

8.2 Progressive Web App (PWA) Farmer Portal (The Field View)

- **Target User:** Smartphone-owning progressive farmers and local village cooperative fleet operators.
- **Key Controls:** Interfaces with active field barrels via local USB serial configurations or short-range wireless configurations. Renders localized visual indicators representing current inner core heat profiles during active pyrolysis, shows total accumulated multi-year personal storage counts, tracks current available financial balances, and features a big button to execute instant Safaricom cash withdrawals.

8.3 Headless 2G Telephony Engine (The Inclusion Layer)

- **Target User:** Vulnerable remote smallholder farmers lacking smartphones and network bandwidth.
- **Key Controls:** Operates entirely over standard cellular lines. Accessible using the USSD directory string *384*55#. The interactive keypad menu reads data states straight out of the core SQLite backend. Farmers check their kiln balances via structured SMS alerts, update active batch sequences, and activate localized automated phone voices that speak instructions directly in Swahili.

9. Comprehensive Monetization & Commercial Architecture

AngaGuard operates on an sustainable, asset-backed **B2B2C Hybrid Ecosystem Software and Transaction Model**. We explicitly avoid charging upfront capital fees to resource-constrained local farmers.



| TOTAL CREDIT SALE VALUE (Carbonmark Sandbox) : \$135.00 |

Farmer Mobile Cashout (M-Pesa) \$50.00 (37.0%)		AngaGuard Take-Rate Fee (Software Platform) \$65.00 (48.2%)	
Cooperative Operational Stipend \$20.00 (14.8%)			

1. **SaaS Infrastructure Tier:** Local SMEs pay a tiered operational monthly platform subscription fee (scaled by transaction lines) to utilize our footprint computation interfaces, recommendation pipelines, and compliance scorecard generators.
2. **The Transaction Take-Rate Stream:** The platform's primary revenue engine runs on an automated programmatic settlement commission docked from international buyers during credit retirement transactions. When a corporate entity or an asset market (Actor 4) clears an AngaGuard carbon credit for **\$135.00** via our mock Carbonmark sandbox pipeline, the revenue split executes automatically:
 - o **The Farmer Cut (\$50.00 equivalent in KSh):** Routed via instant Safaricom B2C mobile financial rails directly to the individual operator's device, delivering a massive secondary income source that shifts their livelihood baseline.
 - o **The Cooperative Cut (\$20.00 equivalent in KSh):** Retained by the local managing hub to fund physical collection logistics, regional site safety oversight, and the continuous return distribution of the generated biochar straight back into rural farming plots.
 - o **The AngaGuard Platform Take-Rate (\$65.00):** Retained by your enterprise to finance data routing nodes, cover cellular shortcode network overhead, maintain hardware tooling units, and scale business expansion.

This structural framework guarantees that every actor's commercial incentives remain aligned with real-world climate adaptation, turning complex digital environmental governance into practical economic prosperity.

We have successfully consolidated the complete engineering blueprint for AngaGuard. Would you like me to write out the **exact 5-minute presentation script** matching this documentation for your pitch to the judging panel, or should we refine the **C++ firmware code** for the WaziDev microcontroller sensor loop?